

Exp-7

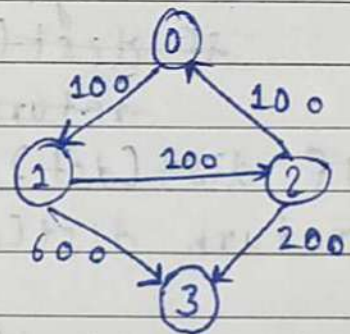
n cities, array flights $flights[i] = [from, to, price]$ indicates that there is a flight from city $from$ to city to with cost $price$.

Given src , dst and K , return the cheapest price from src to dst with at most K stops. If no route -1.

Input: $n=4$, flights = $[[0, 1, 100], [1, 2, 100], [2, 0, 100], [1, 3, 600], [2, 3, 200]]$, $src=0$,

$dst=3$, $K=1$

Output = 700



→ Algorithm :-

- Initialize :- $dist[src] = 0$
 $dist[others] = \infty$
- Repeat $K+1$ times
For every edge (u, v, w)
if $dist[u] + w < dist[v]$:
 $dist[v] = dist[u] + w$
- Return If any edge still relaxes, ~~negative~~ cycle exists.

→ Pseudocode:-

for each vertex:
 $\text{dist}[i] = \infty$;

$\text{dist}[\text{src}] = 0$;

for $i = 0$ to $k-1$:
 $\text{temp} = \text{copy}(\text{dist})$;
 for each edge (u, v, w) in edges:
 if $\text{dist}[u] \neq \infty$ AND $\text{dist}[u] + w < \text{dist}[v]$:
 $\text{temp}[v] = \text{dist}[u] + w$; $\text{min}(\text{temp}[v], \text{dist}[u] + w)$;
 $\text{dist} = \text{temp}$
for each edge (u, v, w) in edges:
 if $\text{dist}[u] \neq \infty$ AND $\text{dist}[u] + w < \text{dist}[v]$:
 return -1;
if $\text{dist}[\text{dst}] == \infty$: return -1;
return $\text{dist}[\text{dst}]$;

→ Time Complexity:- $O(K \times E)$
Space Complexity:- $O(V)$